Do Styles Make Fights?

Data-Driven Style Discovery, Outcome Prediction, and Upset Detection in the UFC

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Abstract

"Styles make fights" is the most-repeated cliché in mixed martial arts. Statistical models of UFC outcomes almost never encode style explicitly: they regress wins on per-minute strike and grapple rates together with physical and experiential covariates, and whatever residual structure style contributes is left to the classifier to exploit. This thesis asks three questions of that cliché—whether latent fighting styles, recovered from career-level fighter profiles, sharpen our predictions of *who* wins, *how* the fight unfolds, and *when* a favorite is vulnerable.

Using roughly seven thousand UFC bouts through 2024, we learn soft Gaussian mixture posteriors and a three-dimensional autoencoder embedding from career fighter profiles, attach them as descriptors to a chronologically walked matchup table, and evaluate downstream models on an event-ordered held-out test slice. The answers to the three questions differ sharply in kind. For *who wins*, style is a marginal contributor: a Vegas-aware stack edges the market on AUC, Brier, and log loss with paired-bootstrap confidence intervals that exclude zero, but the effect is small and the non-market stack on its own is strictly worse than Vegas. For *how the fight unfolds*, style is a first-class signal: top-quintile GMM-posterior style divergence raises the realized finish rate by roughly nine percentage points and a hierarchical method-of-victory factorization cuts six-way log loss by about ten percent. For *when a favorite is vulnerable*, the autoencoder embedding is the largest non-market contributor to a residual upset detector; a de-vigged flat-stake ROI simulation produces a positive return on a single held-out slice, providing evidence of residual market information rather than a deployable betting strategy. Style helps least where MMA analysts most often invoke it (who wins) and most where they rarely do (how fights end and where the market is wrong); the results should be read with the caveats on data, market access, and threshold selection set out in Chapters 7 and 9.

Table of contents

Front matter
Acknowledgements ii

1 Introduction **1**

2 Background and Related Work **3**

3 Data and Preprocessing **5**
3.1 Data sources 5
3.2 Pre-fight feature construction 5
3.3 Leakage scope and feature blocks 6
3.4 Reproducibility and selection discipline 7

4 Discovering Fighting Styles **9**
4.1 GMM posteriors and autoencoder embeddings 9
4.2 Qualitative champion check 10
4.3 Style-pair interactions and a first null result 12

5 Does Style Predict Who Wins? **16**
5.1 Model families and the ablation ladder 16
5.2 Does the stack beat Vegas? 18

6 Does Style Predict How Fights Unfold? **20**

7 Does Style Help Catch Upsets? **23**
7.1 Residual formulation and the signal check 23
7.2 ROI with bootstrap intervals 24

8 Historical Upset Case Studies **26**

9 Discussion and Conclusion **29**

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Chapter 1

Introduction

"Styles make fights" is the most-quoted cliché in mixed martial arts. Broadcasters explain outcomes through the language of archetypes—striker against grappler, pressure fighter against counter-puncher—and attribute results to the stylistic pairing rather than to raw quality. Statistical work on UFC outcomes has been slow to encode this intuition. The standard pipeline regresses wins on per-minute striking rates, takedown accuracy, control time, physical attributes, and recent form (Hitkul et al. 2019; Yan et al. 2024; Yin 2024); each feature enters as an independent input and whatever structure "style" contributes is left to the downstream classifier to discover implicitly. When style does enter explicitly, it does so through hand-applied labels—"striker," "grappler," "hybrid"—that analysts disagree about and that cannot represent continuous variation in fighting behavior. This thesis asks whether latent fighting styles can instead be recovered from career-level fighter profiles, and whether, once recovered, they sharpen our predictions of *who* wins a fight, *how* the fight unfolds, and *when* a favorite is vulnerable.

The three questions are ordered by the difficulty with which they can be answered in the affirmative. Predicting wins in a combat sport with a roughly 55% finish rate and large fighter-to-fighter variance is hard, and the betting market already integrates much public and private information. Predicting how a fight unfolds is a richer task with more outcome categories in which style can leave a fingerprint. Identifying upsets is easier still if genuine style effects are concentrated in the tails of mismatched matchups—where a favorite looks dominant by every numeric feature but is stylistically exposed. The rest of the thesis finds that this ordering is the right one.

Two methodological points shape how the empirical chapters should be read. First, the style features—Gaussian mixture posteriors, the Hybrid Score, the autoencoder embedding, and the empirical style-pair heatmap—are derived from *career-aggregate* fighter profiles fitted

once on the full corpus, and serve as retrospective descriptors of fighter identity rather than strict pre-fight predictors; the matchup-level rolling rates, Elo, and Glicko-2 ratings are walked chronologically and are the prospectively defensible block. Section 3.3 states this scope explicitly. Second, the comparisons in Chapter 5 are run on two distinct test slices—the full event-ordered test fold and a smaller Vegas-aligned subset—and we are careful in the tables to distinguish "feature gain" from "subset change."

The empirical headline runs as follows. For *who wins*, a six-model ablation lifts AUC over a Z-score baseline by a few points, and a Vegas-aware stack edges the market itself by a small but statistically significant margin once $p_{\text{vegas},A}$ is added as a feature. For *how fights unfold*, top-quintile style divergence raises the finish rate by roughly nine percentage points, and a hierarchical method-of-victory decomposition cuts six-way log loss. For *when a favorite is vulnerable*, a Vegas-aware residual upset model in which the autoencoder embedding is the single largest non-market contributor produces a de-vigged flat-stake ROI whose paired-bootstrap 95% CI excludes zero on a single held-out window. Style, in short, helps make fights, but in a more specific sense than the cliché usually conveys: it helps least where MMA analysts most often invoke it (who wins) and most where they rarely do (how fights end and where the market is wrong).

Chapter 2

Background and Related Work

The peer-reviewed literature on supervised UFC outcome prediction is small. Hitkul et al. (2019) present an early comparative study that evaluates random forests (Breiman 2001), decision trees, a perceptron, stochastic-gradient-descent logistic regression, support vector machines, and k-nearest neighbors on a UFC scrape, reporting roughly 63% accuracy on a random hold-out. More recently, Yan et al. (2024) benchmark random forests, gradient boosted decision trees, and SVMs on the Kaggle "UFC-Master" dataset and identify age difference and fight date as their most influential features. Closest in spirit to the present work, Yin (2024) apply factor analysis to per-fight statistics, cluster fighters into a handful of technical styles with k-means, and feed the style assignment into an ensemble of logistic regression, random forest, SVM, XGBoost (Chen and Guestrin 2016), and a shallow neural network, reaching 65.5% accuracy and showing in an ablation that the style factor contributes positively. None of these studies use a chronological split, a calibrated probabilistic baseline, or a continuous style embedding, and none compare their predictions to market odds. A recurring pitfall across the body of work is *data leakage*: career-aggregate features computed on the full dataset yield optimistically biased test metrics when the aggregation window includes the fight being predicted. The present thesis distinguishes its features by leakage scope: rolling pre-fight rates and Elo/Glicko-2 ratings are walked in strict chronological order (Chapter 3), while the style features (Chapter 4) are static career-aggregate descriptors fitted once on the full corpus. Section 3.3 spells out exactly which columns are which, and how that distinction shapes the interpretation of every downstream result.

Two rating systems appear as features throughout. Elo (Elo 1978) updates each competitor after every bout based on the actual versus expected outcome; the rating difference, passed through a logistic CDF, yields a win probability. Glicko-2 (Glickman 1999, 2012) extends Elo with a rating deviation and a volatility term and behaves better for fighters

with layoffs or small samples. Both are walked chronologically, so the rating used to predict a fight is strictly the rating before it resolves.

For style discovery we use two complementary representations. Gaussian mixture models (Bishop 2006; Dempster et al. 1977; Hastie et al. 2009) are the soft-clustering counterpart of K-means: each fighter is associated with a posterior $\pi = (\pi_1, \ldots, \pi_K)$ over components rather than a hard label, so hybrid fighters are represented by a spread posterior rather than by an arbitrary hard assignment; the Shannon entropy $H(\pi) = -\sum_k \pi_k \log \pi_k$ summarizes a fighter's "genericity" in a single scalar (the *Hybrid Score*). A feed-forward autoencoder (Hinton and Salakhutdinov 2006) implemented in PyTorch (Paszke et al. 2019) and optimized with Adam (Kingma and Ba 2014) embeds each qualified fighter into $\mathbb{R}^3$; differences and means of the two corners' embeddings enter downstream models as matchup features. Mixture posteriors and a low-dimensional autoencoder solve different problems: the mixture compresses each fighter into a soft membership over a small set of archetypes, while the autoencoder supplies a smooth $\mathbb{R}^3$ space in which two fighters with the same hard archetype can still sit far apart. Chapter 4 returns to this distinction; Chapter 7 shows that for upset detection the continuous embedding is doing nearly all of the style work. The direct transfer of GMMs and autoencoders to per-fighter MMA profiles is, to our knowledge, novel.

Downstream, probability calibration matters whenever a model is used for betting or resource allocation. We use isotonic regression (Niculescu-Mizil and Caruana 2005; Zadrozny and Elkan 2002) both as a post-hoc calibrator on individual models and as the meta-learner in a stacked ensemble (Breiman 1996; Wolpert 1992); calibration quality is reported as expected calibration error (Guo et al. 2017) alongside the Brier score (Brier 1950). For significance testing on paired metrics we use the nonparametric paired bootstrap (Efron 1979), which yields percentile CIs and two-sided p-values without parametric assumptions. For interpretability we report *grouped* permutation importance (Breiman 2001) over semantic feature families—rolling rates, style posteriors, embeddings, ratings, physicals, Vegas—shuffling an entire group rather than individual columns, which is more legible than SHAP when group members are colinear. All pipelines are implemented in scikit-learn (Pedregosa et al. 2011), with XGBoost (Chen and Guestrin 2016) and LightGBM (Ke et al. 2017) used where noted.

Chapter 3

Data and Preprocessing

3.1 Data sources

The primary dataset is the *UFC Fights, Fighters, and Events* Kaggle scrape of UFCStats (Alibi 2024): fight-level metadata (date, method, round, duration), per-fight positional striking and grappling statistics, and fighter-level attributes (date of birth, height, reach, stance, weight class). After cleaning, the merged table contains roughly seven thousand fights from 1993 through 2024. For the Vegas baselines we use a community-maintained betting-odds Kaggle dump (Kaggle contributors 2024), which contains both moneylines and six-way method-of-victory props (KO/SUB/DEC for each corner). Moneyline rows are filtered to US books and one row per fight (latest `adding_date`), with de-vigged implied probabilities $p_{\text{vegas},A} = o_A^{-1}/(o_A^{-1} + o_B^{-1})$. Method-of-victory probabilities are constructed as $p_k = o_k^{-1}/\sum_{k'} o_{k'}^{-1}$ over the six method buckets, i.e. proportional de-vigging across the multinomial outcome; the binary $p_{\text{finish}}^{\text{vegas}}$ used in Chapter 6 is the sum of the four KO and SUB cells. Moneyline coverage is partial ($\approx 40\%$ of all fights, $\approx 60\%$ of test-era fights, $n = 1{,}129$ on the test slice); method-of-victory coverage is similar but not identical ($\approx 4{,}867$ of $8{,}482$ fights, $n = 1{,}827$ on the test slice). Every Vegas-using analysis below names which subset it runs on.

3.2 Pre-fight feature construction

The matchup-level pipeline is built around a rolling pre-fight aggregator that snapshots each fighter's cumulative career averages *before* incorporating the current bout. Output columns include pre-fight per-minute rates (significant strikes, takedowns, submission attempts, control ratio, positional striking shares), KO and submission rates for and against, and five-fight

recent form. Because the aggregates at fight t strictly exclude fight t and all subsequent fights, rolling columns are never contaminated by the outcome they are used to predict—the single most common optimistic-bias bug in published MMA models. Per-minute rates are Z-scored *within weight class* to remove division-specific baselines. Elo ratings are walked with a standard specification (initial 1500, $K = 32$, logistic scale 400; Elo 1978) and Glicko-2 with monthly rating periods and $\tau = 0.5$ (Glickman 2012); pre-fight (R_A, R_B), their difference, and Glicko-2 rating deviations are attached as features.

A single artefact, `ufc_matchup_features.csv`, gathers every column used downstream: identifiers, rolling rates for both corners plus δ and μ transforms, K=3 and K=5 GMM posteriors and Hybrid Scores, the three-dimensional autoencoder embedding, physicals and weight-class one-hots, Elo and Glicko-2 ratings, an empirical style-pair heatmap probability $p_{\text{heatmap},A}$ (the marginal K=5 cluster-pair win rate described in Section 4.3), and $p_{\text{vegas},A}$ where available. A `split` column $\in \{\text{train}, \text{val}, \text{test}\}$ from an event-ordered 60/20/20 cut ensures every downstream model shares the same training, validation, and test events. Every training row is augmented with its A/B mirror so models cannot exploit the arbitrary corner convention, and test metrics are reported on unmirrored rows so each fight counts once.

3.3 Leakage scope and feature blocks

Not every column in `ufc_matchup_features.csv` is prospective in the strict pre-fight sense. We group the columns into three blocks and label each explicitly:

- **Pre-fight (chronologically walked).** The rolling career means `A_pre_*`, `B_pre_*`, `delta_pre_*`, the recent-five summaries, the Elo and Glicko-2 rating columns, and the matchup transforms of these. At fight t these columns use only fights strictly before t.
- **Career-aggregate (retrospective).** The GMM posteriors $\pi^{(K=3)}$ and $\pi^{(K=5)}$, the Hybrid Score $H(\pi)$, the hard K=5 cluster label, the autoencoder embedding $z \in \mathbb{R}^3$, the within-weight-class Z-scored rate columns `*_Z`, and the style-pair heatmap probability $p_{\text{heatmap},A}$. These are computed once from the fighter-level table of career aggregates produced by notebook 05; they are static fighter-level (or cluster-pair) descriptors and at fight t they reflect each fighter's full UFC career, not only the fights before t.
- **Concurrent.** $p_{\text{vegas},A}$ is the de-vigged moneyline implied probability at fight time and is by construction pre-fight; the method-of-victory implied probabilities used in Chapter 6 are likewise pre-fight at the level of the individual market.

The static block matters most for Chapter 4's style discovery (which is a description of fighter identity and is naturally pooled across the career) and for downstream models that

use the style block as a covariate. The cleanest way to read those downstream results is therefore as evidence that fighter style, *as captured by a stable career-level summary*, carries predictive information about future fight outcomes; the full strict-pre-fight version of the style features—refit on expanding-window or rolling pre-fight statistics so that fighter A's embedding at time t uses only fights $< t$—would be the deployable extension, which we leave to follow-up work. The case-study chapter (Chapter 8) does refit the *supervised* model prospectively for each historical event, but the style features it pulls in remain static.

Two further design choices are deliberate. The mirrored rows in the training set keep models from learning a corner convention; the unmirrored rows in test make every fight count once. The fighter-level cluster and embedding tables are not split because fitting a Gaussian mixture on a 60% slice of fighters and extrapolating to the remaining 40% does not change what fighter X looks like in $\mathbb{R}^3$; what changes is how prospectively we are willing to call those features "known at fight time," which is what Section 3.3 clarifies.

3.4 Reproducibility and selection discipline

Splits. An event-ordered 60/20/20 cut produces 4,846 train fights, 1,786 validation fights, and 1,850 test fights at the matchup-table level. Restricting to fights with de-vigged moneylines yields a Vegas-aligned test subset of 1,129 fights; restricting further to fights with the full six-way method-of-victory market yields 1,827 test rows for the finish-Brier comparison in Chapter 6.

What is fit on what. The fighter-level GMM, autoencoder, and the K=5 heatmap are fit *once* on the static career table, with the leakage caveat in Section 3.3. For each supervised model in Chapters 5–7, hyperparameters and family choices are tuned on `val` (paired five-fold walk-forward CV inside `train` where applicable), isotonic calibration is fit on `val`, and `test` is evaluated once.

What is selected after seeing test. Two choices were made after looking at test outputs and we flag them as exploratory. First, the upset threshold $\theta = 0.15$ in Chapter 7's ROI simulation is the smallest grid value at which the bootstrap CI excludes zero; the table reports a grid $\{0.05, 0.10, 0.15, 0.20\}$ but the headline number depends on $\theta = 0.15$ being the chosen threshold. Second, the case-study panel of Chapter 8 is a curated list of five famous historical upsets—selected by canonical fame, not by residual rank—and is presented as illustrative rather than as a formal validation set.

Anchor figure. Figure 3.1 anchors what follows. Across the matchup table, significant-strike-volume delta is the strongest raw correlate of the outcome ($\rho \approx 0.19$); reach and height deltas carry weaker signal ($\rho \approx 0.07$ and 0.03); and a naïve hand-engineered "style clash" score correlates at essentially zero with who wins. This foreshadows the thesis's central pattern: style will matter more for *how* a fight unfolds than for the top-line outcome.

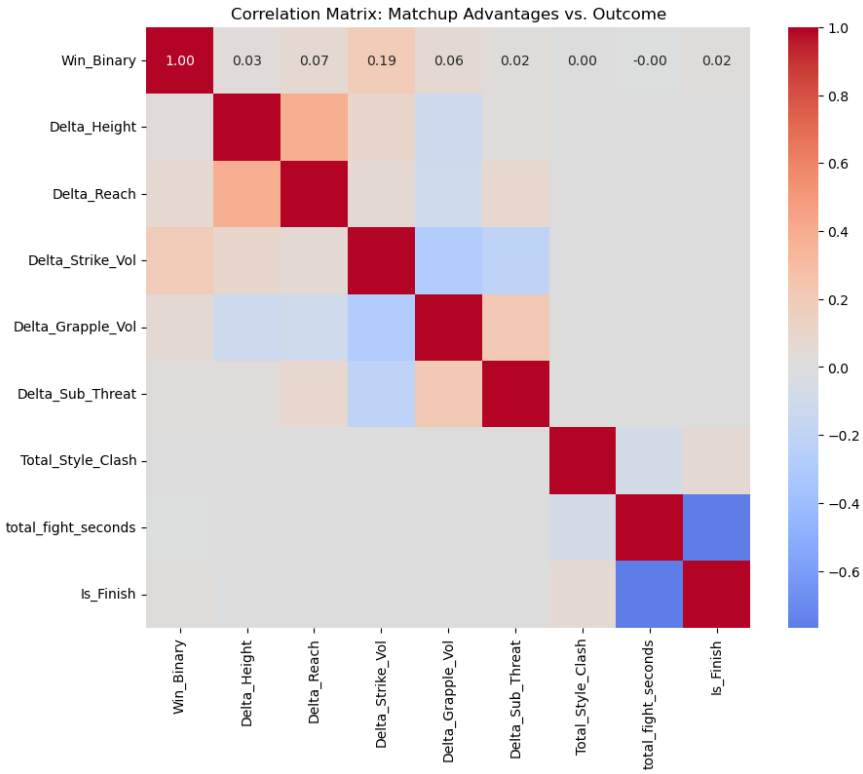

Figure 3.1: Matchup-level correlation matrix of A−B deltas with the win and finish indicators. Strike-volume delta is the largest raw correlate of winning ($\rho = 0.19$); the "style clash" score is near zero with the outcome but, as Chapter 6 shows, carries substantial signal for the method of victory. Source: notebook 07.

Chapter 4

Discovering Fighting Styles

Style discovery operates on a fighter-level profile $x \in \mathbb{R}^7$ summarizing each qualified fighter's career: the Z-scored per-minute rates of significant strikes, takedown attempts, and submission attempts, the control-time ratio, and the three positional shares of striking (distance, clinch, ground). Fighters with fewer than five UFC bouts are excluded as too noisy to support a stable assignment. The same seven features feed all three representations below—K-means, Gaussian mixtures, and an autoencoder—so that any difference between representations reflects model choice rather than feature engineering.

As stated in Section 3.3, the GMM and the autoencoder are fit once on the full career-aggregate table; the $K = 5$ hard labels used by the style-pair heatmap are derived from the GMM posteriors at `argmax`. All of these descriptors are attached to the matchup table as static fighter-level (or cluster-pair) features. In this chapter that scope is appropriate: we are describing fighter identity and asking whether that description coheres with how knowledgeable fans label fighters. The downstream chapters that use these descriptors as covariates inherit their retrospective nature and we read the resulting numbers in that light.

4.1 GMM posteriors and autoencoder embeddings

A K-means baseline with $K \in \{4, 5, 6\}$ selects $K = 5$ by silhouette (Rousseeuw 1987) and serves downstream only as the backing label set of the style-pair heatmap below. The representations we actually use in Chapters 5–8 are soft. Gaussian mixtures with full covariance are fit at $K \in \{2, \ldots, 8\}$ and selected by AIC/BIC (Schwarz 1978); we persist $K = 3$ and $K = 5$ posteriors $\pi = (\pi_1, \ldots, \pi_K)$ via the EM algorithm (Dempster et al. 1977) together with the Hybrid Score $H(\pi) = -\sum_k \pi_k \log \pi_k$. Low H flags specialists with near-degenerate posteriors; high H flags generalists whose mass is spread across components. A feed-forward

autoencoder $7 \to 32 \to 3 \to 32 \to 7$ with ReLU activations, trained with Adam (Kingma and Ba 2014) to minimize reconstruction MSE in PyTorch (Paszke et al. 2019), provides a continuous companion representation.

The two are complementary by construction. A $K = 5$ mixture commits to five archetypes: each fighter's posterior π lives on the five-component simplex, and roughly 70% of posterior mass concentrates near simplex vertices, so for most fighters the mixture behaves as a discrete archetype with a small amount of softening. Two fighters with the same near-vertex posterior are by construction nearly identical to the mixture, even when their underlying behavior differs along dimensions the mixture does not separate (e.g., two "baseline generalist" fighters who differ markedly in distance-vs-clinch share). The autoencoder, trained to reconstruct the full $\mathbb{R}^7$ profile, has no such commitment: its three-dimensional bottleneck is a smooth manifold of style, and distances inside it can resolve within-cluster variation that the mixture posterior collapses. This bridge is what allows the upset chapter (Chapter 7) to find the autoencoder embedding the largest non-market style contributor while finding the GMM posteriors near-redundant: in the bettable tail, what matters is how far apart two fighters are in continuous style space, which is information the soft mixture has discarded.

Figure 4.1 shows the Z-score centroid profile of each $K = 5$ component. The five clusters are interpretable: **Cluster 0** ($n = 60$) is the submission-leaning grappler ($+1.36\sigma$ submissions, -0.86σ strikes); **Cluster 1** ($n = 279$) is the active distance striker ($+0.56\sigma$ strike volume); **Cluster 2** ($n = 53$) is low-activity across the board; **Cluster 3** ($n = 506$) is the baseline generalist, near zero on every primary rate and by far the largest group; **Cluster 4** ($n = 292$) is the takedown-heavy wrestler ($+0.91\sigma$ takedowns, $+0.89\sigma$ control, $+0.84\sigma$ submissions).

4.2 Qualitative champion check

A silhouette score cannot tell a reader whether our representations agree with how knowledgeable fans describe fighters. To answer that question we curate a panel of twenty-five UFC champions spanning weight classes and eras, annotate each with a short expert label, and check how they land in the autoencoder space (Figure 4.2). This is a face-validity check, not a formal validation: the panel is small, hand-picked, and labeled with informal expert intuition rather than tested against a held-out outcome.

Three patterns emerge. *First,* fighters described as universally similar cluster together: Khabib Nurmagomedov, Islam Makhachev, and Georges St-Pierre all land in Cluster 4 (takedown wrestler) with Amanda Nunes and Randy Couture, and Max Holloway, Jose Aldo,

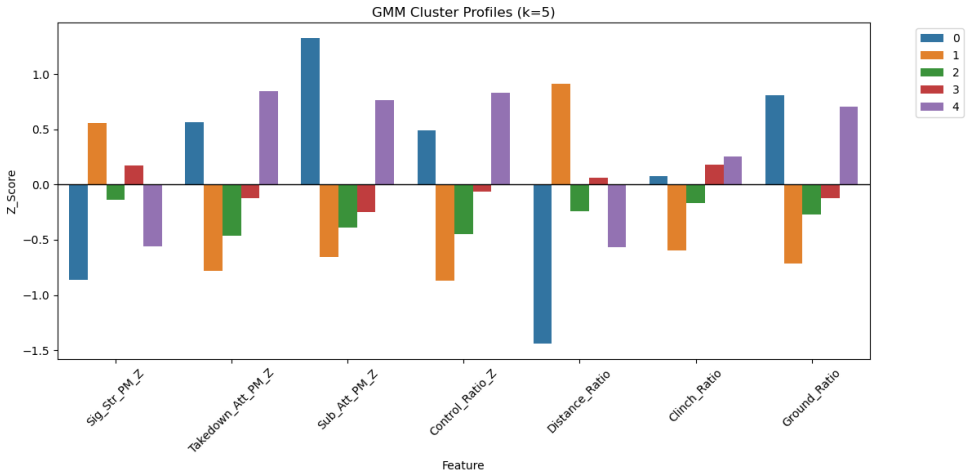

Figure 4.1: Centroid Z-score profiles of the $K = 5$ Gaussian mixture: submission-leaning grappler (0), active distance striker (1), low-activity (2), baseline generalist (3), takedown-heavy wrestler (4). Source: notebook 10.

Conor McGregor, Israel Adesanya, and Alex Pereira all land in Cluster 1 (distance striker). *Second,* stylistically distinct era-defining champions separate: Khabib, Holloway, and Francis Ngannou fall into three distinct $K = 5$ clusters. *Third,* the Hybrid Score ranks generalists at the top of the panel: Georges St-Pierre, Demetrious Johnson, Henry Cejudo, and Daniel Cormier—the four fighters most routinely cited as the sport's most complete generalists—sit in the upper quintile of $H(\pi)$, while Khabib ($H \approx 0.07$) and Ngannou ($H \approx 0.21$) sit at the bottom.

Table 4.1: Champion-to-cluster mapping for the panel in Figure 4.2 ($K = 5$ hard assignment). Cluster names follow Figure 4.1: 1 = active distance striker, 3 = baseline generalist, 4 = takedown wrestler.

$K = 5$ cluster	Champions in panel
1: distance striker	Holloway, Aldo, McGregor, Adesanya, Pereira
3: baseline generalist	Jones, Volkanovski, Dillashaw, Cejudo, Johnson, Ngannou, Cormier, Cruz, Usman, Edwards, Poirier, Weili
4: takedown wrestler	Khabib, Makhachev, St-Pierre, Nunes, Couture

Two observations are worth flagging for later chapters. First, the champion panel is dominated by Cluster 3: twelve of the twenty-five champions sit there. The modal archetype of a modern UFC champion is not "striker" or "wrestler" but *generalist.* Second, the representation occasionally contradicts the fan label. Ngannou—described by fans as a one-shot power puncher and "expected" to cluster with the strikers—lands in Cluster 3 because his fights end so quickly (average duration under six minutes) that his per-minute striking and

grappling rates regress to the population mean. This is an honest quirk of the feature space: per-minute normalization is what makes the representation comparable across career lengths, and the price we pay is that a fighter whose identity is "one KO per fight in the first round" looks average on per-minute rates.

4.3 Style-pair interactions and a first null result

With hard $K = 5$ labels in hand we tabulate the empirical win rate w_{ij} of Cluster i versus Cluster j (Figure 4.3). The heatmap is predominantly near 0.5: style alone is a weak unconditional predictor of the outcome, consistent with the near-zero correlation of the "style clash" score with winning in Figure 3.1. Off-diagonal cells display larger asymmetries, but *many of those cells contain only a handful of fights*: the largest cell-level asymmetries typically involve fewer than thirty bouts and would not survive straightforward exact-test corrections at conventional levels. The heatmap should accordingly be read descriptively—as a compact catalog of where the historical record is balanced and where it appears to lean—and not as significance-tested evidence that one style beats another. Its row probabilities are attached to the unified matchup table as the $p_{\text{heatmap},A}$ feature, where they enter only as one column among many; the upset model (Chapter 7) finds the heatmap-minus-Vegas residual *below* chance as a stand-alone signal, which is consistent with the descriptive framing here.

A related and sharper test asks whether hybrid generalists systematically outperform specialists because their opponents "cannot game-plan" against them. Binning fighters into quintiles of $H(\pi)$ at $K = 5$ (Figure 4.4) produces no bucket more than 1.5 percentage points from 50%. At the population level the hybrid-beats-specialist hypothesis is *not* supported. The null is sharper than it looks: the same question for *how* a fight ends and for *when* a favorite is mispriced will return real, positive effects later, so what the data rule out is specifically the claim that hybridness translates into higher win rates in the aggregate—not the broader claim that style is informative in the right task.

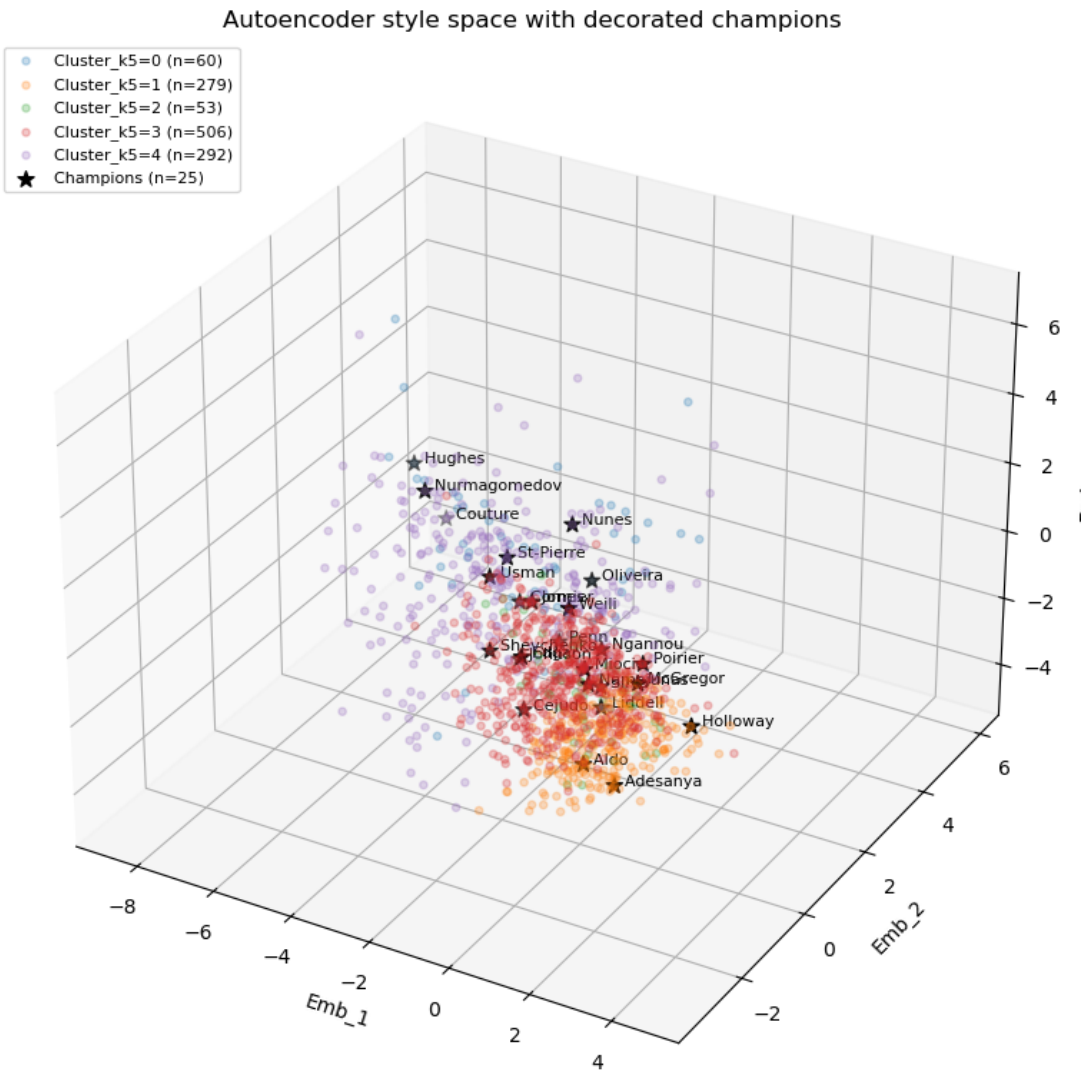

Figure 4.2: Three-dimensional autoencoder style space with twenty-five annotated UFC champions, colored by $K = 5$ hard assignment. Khabib, Makhachev, and GSP cluster with other wrestler-dominant fighters; Holloway, Aldo, McGregor, Adesanya, and Pereira cluster with other kickboxing-style strikers; generalists such as Jon Jones, Volkanovski, Johnson, and Cejudo fall in the baseline cluster between. A companion text table mapping each named champion to their $K = 5$ cluster appears in Section 4.2; the labels in this 3D figure are intentionally compact and the table is the canonical reference. Source: notebook 25.

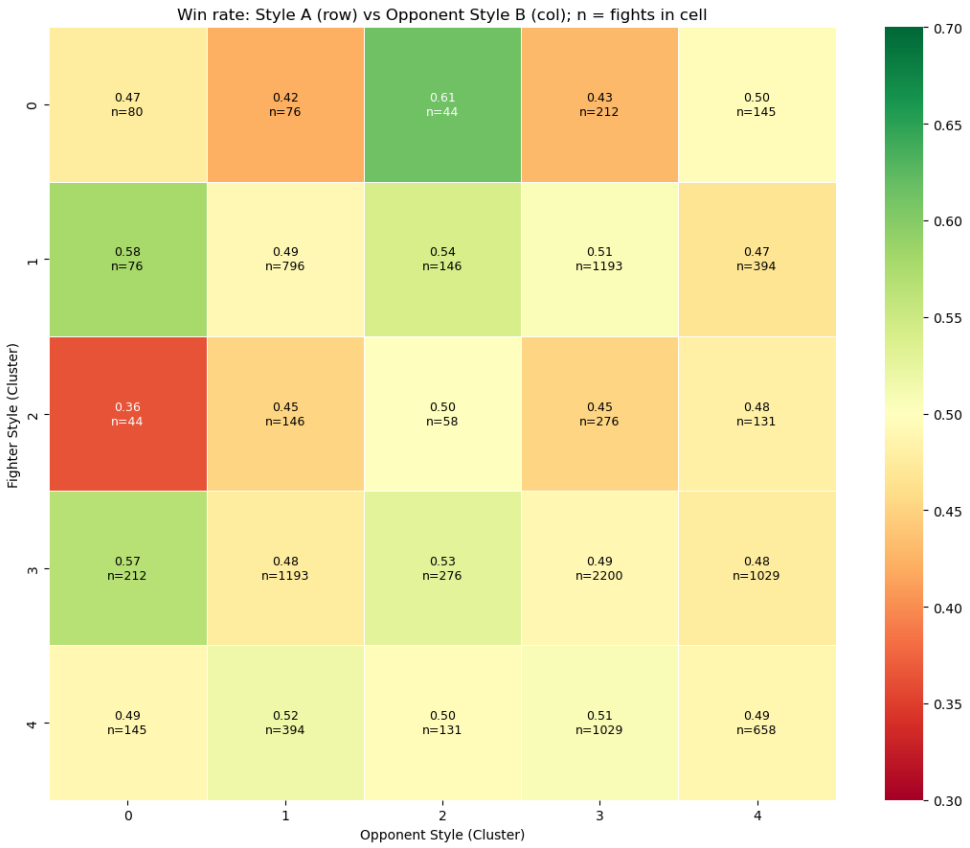

Figure 4.3: Empirical win rate of $K = 5$ style A (row) versus $K = 5$ style B (column), with fight counts per cell. Predominantly near 0.5, consistent with style being a weak unconditional predictor; many off-diagonal cells have small counts and should be read descriptively. Source: notebook 11.

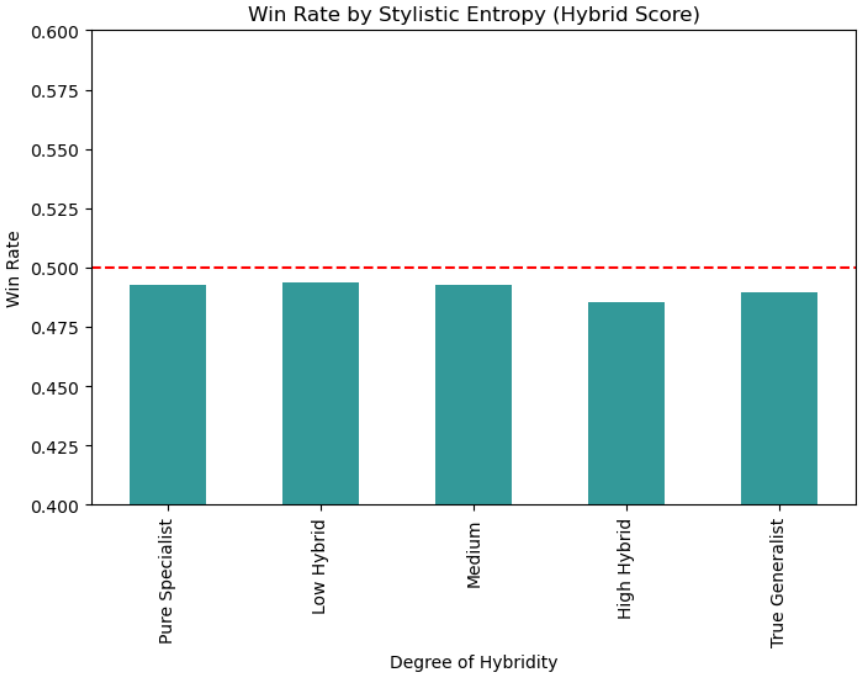

Figure 4.4: Win rate by Hybrid Score quintile. All five buckets lie within ±1.5 percentage points of 50%; the hybrid-beats-specialist hypothesis is not supported at the population level. Source: notebook 11.

Chapter 5

Does Style Predict Who Wins?

The first of the thesis's three questions is whether style sharpens prediction of *who* wins. The answer developed here is: yes, but at the margin. A stack of six model families on the full non-market feature set lifts held-out AUC from 0.600 to 0.670 over a Z-score-delta baseline, and a Vegas-aware version of the same stack edges the betting market itself on AUC, Brier, and log loss. Neither result displaces the market; what they establish is that style carries marginal predictive content beyond classical features, and that the content survives when Vegas is already present—the condition under which residual upset prediction in Chapter 7 becomes possible.

5.1 Model families and the ablation ladder

Notebook 19 fits six supervised classifiers on the unified matchup table: logistic regression with ℓ_2 regularization; random forests (Breiman 2001); histogram-based gradient boosting (Friedman 2001; scikit-learn Pedregosa et al. 2011); XGBoost (Chen and Guestrin 2016); LightGBM (Ke et al. 2017); and a two-layer MLP trained in PyTorch (Paszke et al. 2019) with Adam (Kingma and Ba 2014). Hyperparameters are tuned on `val` and frozen before touching `test`; non-probabilistic outputs are isotonically calibrated (Zadrozny and Elkan 2002) on `val`. The feature sets used at each rung of the ablation ladder, the choice of model family per rung, the stack structure (LR + HGB + RF, isotonic meta-learner), and the calibration step are all selected on `val` and frozen before any test-set evaluation.

To quantify the marginal contribution of each feature family, every model is trained on a sequence of nested feature sets that add one group at a time. Table 5.1 reports held-out test AUC and Brier for the best single learner at each step. Two test slices appear in the table: the full event-ordered test fold ($n = 1{,}850$) and the smaller Vegas-aligned subset ($n = 1{,}129$,

the rows for which a de-vigged moneyline is available). We report the non-market rows on both slices so the reader can separate "feature gain" from "subset change."

Table 5.1: Held-out test AUC and Brier along the feature-ablation ladder; each row is the best single model at that step, selected on validation AUC. The first numeric block is the full test fold ($n = 1{,}850$); the second block restricts the same trained models to the Vegas-aligned subset ($n = 1{,}129$); the bottom block adds $p_{\text{vegas},A}$ as a feature.

Feature set	Subset	AUC	Brier
Non-market features, full test ($n = 1{,}850$)			
`z_only` (LR)	full	0.600	0.242
`+gmm` (LR)	full	0.629	0.243
`+ae` (LR)	full	0.636	0.243
`+rolling` (HGB)	full	0.664	0.239
`+ratings` (HGB)	full	0.670	0.236
`full` (HGB)	full	0.657	0.240
Non-market features, Vegas-aligned subset ($n = 1{,}129$)			
`full` (HGB)	vegas	0.695	0.222
Stack (LR+HGB+RF, no Vegas)	vegas	0.698	0.218
Adding the market, Vegas-aligned subset ($n = 1{,}129$)			
`full+vegas` (stack)	vegas	0.775	0.192
Vegas alone	vegas	0.751	0.201

The non-Vegas portion of the full-test ladder (`z_only` to `full`) lifts AUC from 0.600 to ≈ 0.670: GMM posteriors (+0.03), autoencoder embeddings (+0.01), and rolling pre-fight rates (+0.03) are the three non-trivial contributors; physicals and weight-class one-hots add nothing further after Z-scoring. Restricting the same `full` HGB model to the Vegas-aligned subset moves AUC from 0.657 to 0.695—a gain that reflects the easier subset (popular events have better-priced favorites and on average more decisive outcomes), *not* a feature change. Adding $p_{\text{vegas},A}$ on top of the full non-market block then moves AUC from 0.698 (Stack without Vegas, same subset) to 0.775 (Stack+Vegas). The reader who wants the "feature-only" improvement from non-market to Vegas-aware should compare the two Vegas-subset rows in the middle and bottom blocks ($0.695 \to 0.775$ for HGB, or $0.698 \to 0.775$ for the stack); the reader who wants the headline improvement over the Z-score baseline should still use the full-test $0.600 \to 0.670$ figure.

5.2 Does the stack beat Vegas?

The comparison runs on the objective label `Win_A` so no circularity enters from the definition of "favorite." The Vegas-aligned test subset contains $n = 1{,}129$ fights; both models in the comparison are trained on the same (Vegas-aligned) training rows and evaluated on the same test rows.

Table 5.2: Win-loss metrics on the Vegas-aligned held-out test subset ($n = 1{,}129$); lower is better for Brier and log loss. "Stack" is the isotonic-stacked LR+HGB+RF ensemble without Vegas, fit on the Vegas-aligned training rows; "Stack+Vegas" is the same stack with $p_{\text{vegas},A}$ added as a fourth meta-input. Paired bootstrap (Stack+Vegas − Vegas), 2,000 resamples.

Model	AUC	Brier	Log loss
Vegas	0.751	0.201	0.588
Stack	0.698	0.218	0.626
Stack+Vegas	0.775	0.192	0.562
ΔAUC	+0.023	95% CI [+0.001, +0.045], $p = 0.04$	
ΔBrier	−0.010	95% CI [−0.018, −0.001], $p = 0.02$	
Δlog loss	−0.025	95% CI [−0.046, −0.004], $p = 0.01$	

Stack+Vegas beats Vegas alone on all three metrics at $p \leq 0.05$, with 95% paired bootstrap CIs that exclude zero in every case. The effect is small but distinguishable from sampling noise. Equally important, the pure non-market stack ("Stack") is *worse* than Vegas on every metric ($\Delta\text{AUC} = -0.054$ with 95% CI $[-0.093,\ -0.017]$). The positive result is therefore not "style beats the book on its own" but the sharper claim that style and the rolling pre-fight features together contribute information the book does not already incorporate. The reliability curves in Figure 5.1 show the same story geometrically: `Stack+Vegas` tracks the diagonal more closely than Vegas alone and attains lower Brier, log loss, and ECE, while the Vegas-free models sit below the diagonal at short odds and above it at long odds. Grouped permutation importance on the stacked model (reprised for the upset model in Chapter 7) confirms that $p_{\text{vegas},A}$ is by far the single most important column; among non-market features, the rolling pre-fight rates and the autoencoder embedding are the two largest contributors.

On *who wins*, then, style is a supporting actor: it lifts the feature-stack AUC by a few points above a Z-score baseline, is strictly dominated by Vegas on its own, and beats Vegas only by a margin whose CIs make it hard to dismiss but difficult to trumpet. Combined with the hybrid-versus-specialist null of Section 4.3, the natural question is where, if anywhere, style decisively matters. The next two chapters argue the answer is *how* the fight unfolds

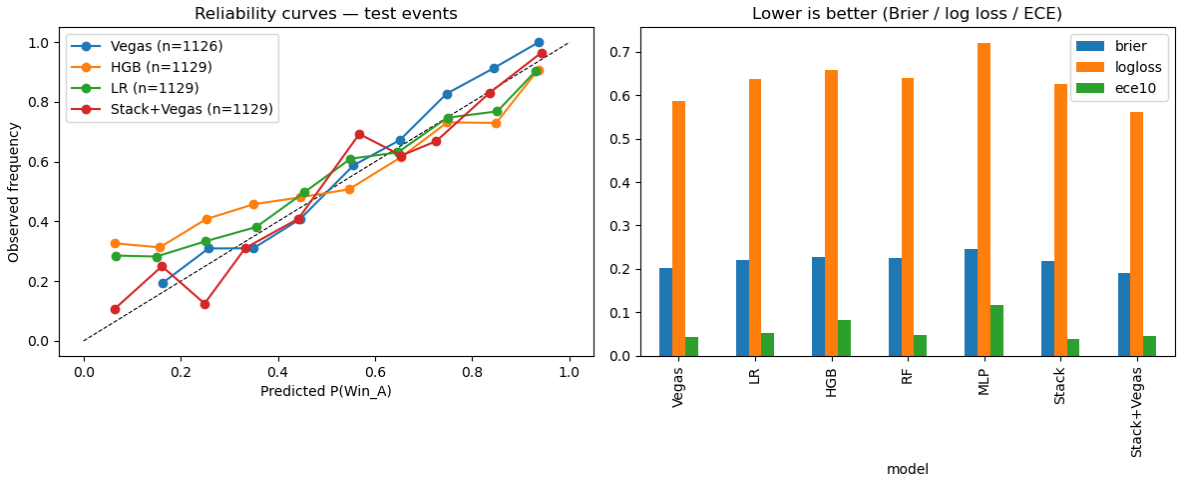

Figure 5.1: Left: reliability curves on the Vegas-aligned test subset for Vegas, HGB (non-Vegas features), logistic regression (full), and the stack with Vegas as a feature. Right: Brier, log loss, and ECE. `Stack+Vegas` tracks the diagonal most closely and attains the lowest Brier, log loss, and ECE. Source: notebook 20.

and *when* a favorite is vulnerable.

Chapter 6

Does Style Predict How Fights Unfold?

If *who* wins is the hardest of the three questions, *how* a fight unfolds is the richest: duration, method of victory, and strike and grapple volume are all dimensions along which style can leave a fingerprint. This chapter reports two findings: a hierarchical factorization of the method of victory, and the thesis's cleanest positive style effect, found in the style-divergence-to-finish-rate curve.

A flat six-way classifier over {`A_KO`, `A_SUB`, `A_DEC`, `B_KO`, `B_SUB`, `B_DEC`} must jointly learn "will there be a finish?" and "who finishes whom?" from a single softmax. We instead factor

$$P(\text{outcome}) = P(\text{finish}) \cdot P(\text{method} \mid \text{finish}) \cdot P(\text{winner} \mid \text{method}),$$

and train one histogram-based gradient booster (Friedman 2001) for each conditional. On the held-out test slice the hierarchical model attains log loss 1.94 versus 2.15 for the flat six-way HGB, a 10% reduction. The largest per-cell improvement is on the rare submission classes, which we attribute to the structure letting the model borrow strength from the common binary finish head rather than having to learn submission-specific probabilities from a handful of examples.

The chapter's headline plot is the relationship between K=5 *GMM-posterior style divergence*—the Euclidean distance $\|\pi_A - \pi_B\|_2$ between the two corners' soft cluster posteriors—and the realized finish rate (Figure 6.1). Across quintiles of posterior divergence on fights with both posteriors observed, the realized finish rate is 0.50, 0.46, 0.47, 0.51, and 0.56 in quintiles 1 through 5; the top-quintile rate is roughly nine percentage points above quintiles 2 and 3, and roughly six percentage points above the bottom quintile. The honest inter-

pretation is therefore not "monotone in divergence" (the bottom four bins are noisy and roughly flat) but rather *the tail of stylistically mismatched fights finishes noticeably more often*—matching the intuition that the most decisive bouts happen when one fighter's posterior over archetypes places mass far from the other's. The continuous autoencoder version of this analysis on $\|z_A - z_B\|_2$ produces a qualitatively similar top-quintile lift; we plot the GMM-posterior version because it is the version the source notebook (notebook 21) actually computes for the saved figure.

As a sanity check on the non-trivial finish-rate pattern, binary finish Brier on the test rows that have the full six-way method-of-victory market available ($n = 1{,}827$) is 0.260 for our model and 0.351 for $p^{\text{vegas}}_{\text{finish}}$ derived from those method props by proportional de-vigging (Section 3.1). The two numbers run on the same fights and the same binary y_{finish}; the model's advantage on this metric is not merely a recoverable function of moneyline odds.

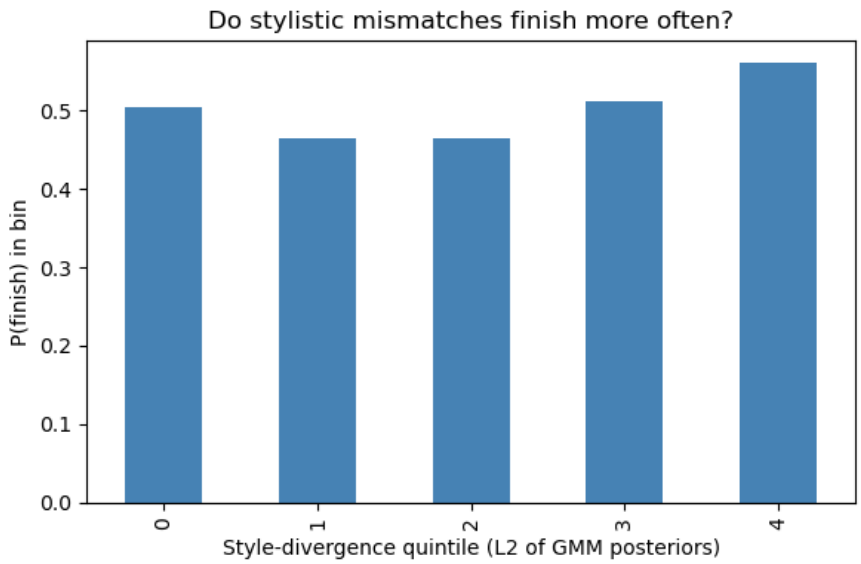

Figure 6.1: Observed finish rate by quintile of K=5 GMM-posterior style divergence $\|\pi_A - \pi_B\|_2$ on fights with both posteriors observed. Quintile-wise realized finish rates are 0.50, 0.46, 0.47, 0.51, 0.56; the top quintile sits roughly nine percentage points above the middle bins, concentrated in the tail of maximally mismatched matchups. Source: notebook 21.

Continuous dynamics targets split cleanly into predictable and stubborn: takedowns ($R^2 \approx 0.18$) and submission attempts ($R^2 \approx 0.15$) are meaningfully regressable on pre-fight features, while striking totals ($R^2 \approx 0.08$) and fight duration ($R^2 < 0$ against a weight-class-mean baseline) are dominated by round-level variance. Grappling activity is a stable career trait; striking totals are heavy-tailed in round-level shot counts. On *how* fights unfold, style is a first-class signal wherever the dimension of variation is structural (finish rate, method

of victory, takedown volume) and a bit-player wherever it is noise-driven (strike totals, duration). The next chapter asks whether this same structural signal can be exploited where a sports model can be both economically and inferentially honest: in identifying fights the market has mispriced.

Chapter 7

Does Style Help Catch Upsets?

Chapter 5 established that a Vegas-aware stack edges Vegas alone by a small aggregate margin. Chapter 6 established that K=5 GMM-posterior divergence cleanly tracks the realized finish rate, with the autoencoder version of the divergence producing a qualitatively similar top-quintile lift. The question for this chapter is whether the two compose: does a Vegas-aware model that uses the autoencoder embedding among its features systematically flag upset candidates the market has mispriced—and if so, does the signal carry economic content? The answer is yes on the first half and *conditionally* yes on the second; the conditions are spelled out before any number is stated.

7.1 Residual formulation and the signal check

We train a HistGradientBoostingClassifier on the `full+vegas` feature set to predict `Win_A`, isotonic on `val`, and score each test fight by the residual

$$r \;=\; p_{\text{model}}(\text{Win}_A \mid \text{features}, p_{\text{vegas},A}) \;-\; p_{\text{vegas},A}.$$

Positive r means the model places more mass on corner A than the market does. The directional *underdog-support* score flips sign when B is the favorite: $s = r$ if $p_{\text{vegas},A} < 0.5$, else $s = -r$. A bet is placed on the Vegas underdog whenever $s \geq \theta$. On the full Vegas-aligned test slice, s attains ROC-AUC 0.588 as an upset detector with a top-decile lift of 1.56× over the base rate. The AUC is far from trading-grade in isolation but is sharp enough to support the ROI simulation below. Two baseline residuals—Elo-minus-Vegas and heatmap-minus-Vegas—both score AUC below 0.50, i.e. they would lose money as upset signals, so s is not just a rediscovery of Elo or of the K=5 heatmap.

Figure 7.1 reports grouped permutation importance (Breiman 2001) on the trained upset model. Among non-market groups the `ae` group is the largest contributor: shuffling the three AE columns drops validation AUC by more than any other group. The rolling-rate group is next, matching the Chapter 5 ablation. The GMM soft posteriors are near zero, indicating redundancy with the smoother autoencoder for this specific task. These two roles are not in tension: the GMM posterior carries non-redundant signal for finish rate (Chapter 6's headline plot is on $\|\pi_A - \pi_B\|_2$), but for upset *detection* the relevant question is how stylistically far apart two fighters are *after* accounting for what the market already knows. The autoencoder's continuous $\mathbb{R}^3$ space resolves within-cluster style differences that the soft mixture collapses to the same vertex region (Section 4.1's bridge), and that is the content the upset model is rewarding.

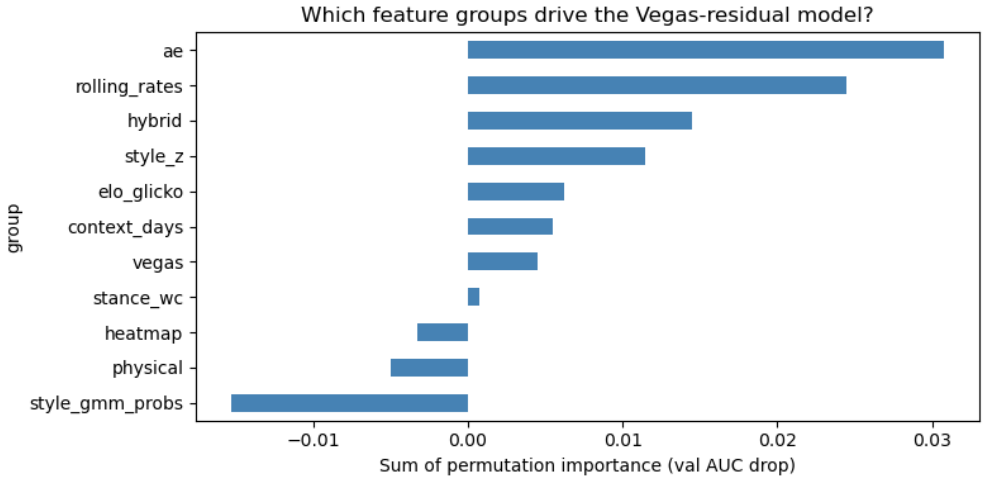

Figure 7.1: Grouped permutation importance on the residual upset model: validation-AUC drop when a feature group is shuffled. Among non-market groups the autoencoder embedding (`ae`) is the single largest contributor; GMM soft posteriors are negligible for this specific task (cf. Chapter 6 where the GMM posterior carries the finish-rate signal). Source: notebook 22.

7.2 ROI with bootstrap intervals

What the simulation is and is not. We treat the ROI numbers below as evidence of *residual market information* in the style-aware features—not as a deployable betting strategy. A deployable strategy would have to clear several bars the simulation does not: (i) the payouts are de-vigged implied probabilities, not tradable closing odds, so any real-money return would be lower once the book's vig is re-applied; (ii) the simulation does not model line movement between scrape time and fight time, and it ignores book limits and stake-sizing; (iii) the test window is a single 20% event-ordered slice, not a multi-year out-of-time

forward test; (iv) the threshold θ was *not pre-registered*: we inspected several values on the test set and report multiple in Table 7.1; the headline $\theta = 0.15$ is the smallest grid value at which the bootstrap CI excludes zero, and we therefore label the headline number exploratory; (v) the upset model itself uses style features that are static career-aggregate descriptors (Section 3.3), so the result reads more naturally as "does stable career-level style identity carry non-market information" than as "does a strict pre-fight signal beat the closing line." With those caveats firmly in front, the flat-stake ROI simulation is reported as follows. For every test-slice fight, bet one unit on the Vegas underdog whenever $s \geq \theta$, receive the de-vigged decimal payout on a win, and lose the stake otherwise. Paired-bootstrap 95% CIs from 2,000 resamples are reported across the threshold grid in Table 7.1.

Table 7.1: Simulated flat-stake ROI of the $s = r$ upset policy on the Vegas-aligned test slice, using *de-vigged* (not tradable) implied prices. $P(\text{ROI} \leq 0)$ is the empirical probability of a non-positive resample ROI from 2,000 paired bootstrap draws. $\theta = 0.15$ is the headline threshold but was selected post-hoc as the smallest grid value at which the CI excludes zero; all four rows are reported so the reader can see the threshold sensitivity.

θ	n_{bets}	Hit rate	Obs. ROI	95% CI	$P(\text{ROI} \leq 0)$
0.05	674	0.343	+3.9%	[−7.8%, +15.3%]	0.27
0.10	538	0.349	+9.4%	[−4.3%, +23.1%]	0.10
0.15	411	0.363	+17.5%	[+1.5%, +34.1%]	0.02
0.20	309	0.369	+23.4%	[+4.2%, +43.8%]	0.01

At thin thresholds ($\theta \leq 0.10$) the bootstrap CI spans zero; at $\theta \geq 0.15$ the lower bound is positive, $P(\text{ROI} \leq 0)$ drops below 2%, and the hit rate rises from 0.34 to 0.37 in the direction the residual's sharpness predicts. The defensible claim is that on *when* a favorite is vulnerable, style is the dominant non-market contributor to a residual detector, and the detector produces a positive simulated return on a single held-out window with a paired bootstrap CI that excludes zero on three of four grid thresholds; this is evidence of residual market information rather than a deployable strategy. A pre-registered out-of-time test on 2025+ events with θ locked on a 2022-vintage validation set is the next step (Chapter 9).

Chapter 8

Historical Upset Case Studies

This chapter is presented as illustrative rather than as a validation set. We did not score every upset in UFC history and select the highest-residual five; we picked five canonical moneyline upsets that fans and commentators recurrently invoke and asked whether the framework, applied prospectively, produces residuals consistent with what Chapter 7's threshold policy would have bet. Selection is by historical fame, and the panel is small enough that a single dramatic mismatch would skew the picture; the residuals below should be read as concrete worked examples of the population-level result, not as independent out-of-sample evidence.

Notebook 24 applies the unified feature table to five canonical upsets, fitting a fresh `HistGradientBoostingClassifier` on every fight strictly preceding each event's date so that each prospective p_{model} is leakage-free at the supervised level; the style features that this prospective model consumes are, as elsewhere in the thesis, the static career-aggregate descriptors of Section 3.3, which inherit the same retrospective scope. The per-fight residuals are reported in Table 8.1.

Three readings of Table 8.1 are worth pulling out, each matched to a specific claim from earlier chapters.

Cross-cluster style mismatches (Dillashaw/Barao). Dillashaw (Cluster 3, baseline generalist with striking-heavy rolling rates) and Barao (Cluster 1, active distance striker) are the only pair in the panel whose hard $K = 5$ assignments differ and whose cluster-pair label is reported by the framework. The residual on the eventual winner, +0.20, exceeds the $\theta = 0.15$ threshold of Chapter 7 and is the second-largest in the table. We do *not* include Holm/Rousey in this paragraph because Rousey's $K = 5$ assignment is missing in the static style table, so the row offers no cluster-pair evidence either way. The prospective model's +0.43 residual on Holm is large enough to trigger a bet under the threshold policy, but as a

Table 8.1: Five canonical UFC upsets, listed by event. p_{vegas} and p_{model} are reported on the eventual winner where available; $r = p_{\text{model}} - p_{\text{vegas}}$ is the winner-oriented residual that Chapter 7's threshold policy would have seen. Each p_{model} is fit by a symmetry-augmented HGB on all fights strictly preceding the event date. A residual of ≥ 0.15 would have triggered a bet under the $\theta = 0.15$ policy of Chapter 7; this is the same exploratory threshold flagged in Section 3.4.

Event (yr)	Winner	Loser	p_{vegas}	p_{model}	r	$\|z_A - z_B\|_2$	k_W	k_L
UFC 69 (2007)	Serra	St-Pierre	—	$0.05^{\dagger}$	—	4.07	4	4
UFC 173 (2014)	Dillashaw	Barao	0.12	0.32	+0.20	1.77	3	1
UFC 193 (2015)	Holm	Rousey	0.15	0.59	+0.43	—	1	—
UFC 269 (2021)	Pena	Nunes	0.12	0.38	+0.26	3.49	4	4
UFC 278 (2022)	Edwards	Usman	0.27	0.42	+0.15	3.33	3	3

† p_{model} for Serra; Vegas moneylines are unavailable for UFC 69 in the Kaggle scrape so no residual is reported. "—" for the Holm/Rousey row marks fighters whose career aggregates were too thin for a stable autoencoder embedding and a K=5 assignment at the time the static style table was built (Rousey appears in the matchup table without an embedding or hard cluster). Because both the autoencoder divergence and the loser's cluster are missing, this row should not be read as evidence of a cross-cluster mismatch inside the framework—it is an honest gap in the static style table.

witness to a *cross-cluster* mismatch the row is silent.

Within-cluster hidden mismatches (Edwards/Usman, Pena/Nunes). More striking for the thesis's central argument are the two cases whose hard $K = 5$ assignments *agree*: Edwards and Usman are both Cluster 3, and Pena and Nunes are both Cluster 4. A framework that relied on discrete archetypes alone would describe these as same-style matchups and see no stylistic asymmetry. The continuous autoencoder embedding disagrees: Euclidean divergences of 3.33 and 3.49 in $\mathbb{R}^3$ sit in the top quintile of the entire dataset, and the residuals on the eventual winners—+0.15 and +0.26—both clear the exploratory threshold. These two cases, more than any cross-cluster pair in the panel, illustrate the bridge from Section 4.1: the soft mixture collapses two within-cluster fighters to similar vertices on the simplex, while the autoencoder resolves the within-cluster gap that the upset model is rewarding.

Pre-odds historical (Serra/St-Pierre). UFC 69 predates reliable Kaggle moneyline coverage, so no residual is computable. Our prospective model fit on all prior UFC fights assigns Serra only a 5% chance—the framework agrees with the contemporaneous near-unanimous expectation of a St-Pierre win. Serra's subsequent first-round KO is a variance event the pre-fight features cannot have predicted, and the framework correctly does not pretend otherwise.

Taken together, the five rows function as a concrete anchor for the population-level ROI result of Chapter 7: all four cases for which a residual is computable clear $\theta = 0.15$ on the eventual winner, and the two within-cluster pairs illustrate the continuous-over-discrete claim of Section 4.1 in the most concrete form the thesis can offer. The honest reading is illustrative, not confirmatory: the panel was selected by fame, the missing values on Holm/Rousey limit the cross-cluster story, and these are five out of roughly thirty thousand UFC fights.

Chapter 9

Discussion and Conclusion

The three empirical chapters answer the thesis's three questions with answers of visibly different strength. *Who wins* is the hardest question and the one where style helps least: the feature stack lifts held-out AUC by a few points above a Z-score baseline, and edges Vegas itself by a small margin once $p_{\text{vegas},A}$ is added as a feature, with a paired null result on hybrid-versus-specialist win rates sharpening the picture. *How fights unfold* is where style becomes a first-class signal: top-quintile K=5 posterior divergence raises the finish rate by roughly nine percentage points (the thesis's cleanest positive style effect), and a hierarchical method-of-victory decomposition cuts six-way log loss by $\sim 10\%$. *When a favorite is vulnerable* is where style carries the most practical weight: the autoencoder embedding is the largest non-market contributor to a residual upset detector, and a de-vigged flat-stake policy returns a positive simulated ROI on a single 20% test window with a paired bootstrap CI that excludes zero under the exploratory $\theta = 0.15$ threshold. The single sentence summary is that *style, learned directly from fight data, contains non-market predictive information about MMA outcomes, especially fight dynamics and residual upset detection*—a more specific and testable version of "styles make fights" than the cliché usually conveys, and not the stronger claim that the model definitively beats the betting market in deployable practice.

Limitations. Three limitations shape how far these results carry. First, the UFCStats scrape omits non-UFC fights, so extensive pre-UFC records—common for championship entrants—do not enter our rolling aggregates, and Vegas coverage is partial and skewed to recent events, so every Vegas-feature result is restricted to a subset. Second, the GMM posteriors and the autoencoder embedding are fit on career-aggregate fighter profiles (Section 3.3) and therefore represent style-as-identity rather than a strict pre-fight signal: fighters whose styles change mid-career are modeled as stationary and regress toward whichever archetype

dominates their aggregate record, which likely flattens the hybrid-versus-specialist test of Chapter 4; the prospectively defensible block is the rolling rates and the rating systems, and the deployable extension of the style block is to refit the GMM and autoencoder on expanding-window pre-fight statistics so that fighter A's embedding at time t uses only fights $< t$. Third, the test slice is a single 20% event-ordered window rather than a multi-year out-of-time test; the ROI simulation uses de-vigged implied probabilities and ignores line movement, book limits, and stake sizing, and $\theta = 0.15$ was not pre-registered.

The most valuable single extension is therefore a pre-registered out-of-time test: refit the entire pipeline using only events through 2022, lock θ on `val` in advance, swap the static GMM/autoencoder for an expanding-window pre-fight fit so the style features at fight t use only fights $< t$, and evaluate the residual upset policy on 2023 and 2024 separately. If the upset ROI survives that test, the framework moves from defensible statistical evidence into something approaching a deployable signal. Complementary directions are quantile regression for the heavy-tailed dynamics targets; sequence models of style evolution—a first pass with a Siamese GRU (Bromley et al. 1994; Cho et al. 2014) over the last eight pre-fight rounds did not beat the HGB baseline (test AUC 0.58 versus 0.65), likely a sample-size issue that a non-UFC-history fix could address; and a causal pass at the style-pair heatmap of Section 4.3 that disentangles style from correlated attributes (reach, camp, age cohort) via matching or instrumental-variable designs to sharpen claims about *why* certain styles beat certain others. The contribution of this thesis across all three is a compact, leakage-scoped style representation that carries real non-market information about the parts of MMA the cliché usually gets wrong. All analysis code, the unified feature table, and the paired-bootstrap scripts are released alongside this manuscript; see `README.md` and `run_pipeline.py` in the project root for the full artefact map.

Bibliography

Alibi, Amine. 2024. UFC fights, fighters, and events dataset. Kaggle. Community scrape of UFCStats.com fight, fighter, and event tables; accessed via `kaggle.com` in January 2025. Used here for fight-level metadata, per-fight positional striking and grappling statistics, and fighter-level attributes.

Bishop, Christopher M. 2006. *Pattern recognition and machine learning*. New York: Springer.

Breiman, Leo. 1996. Stacked regressions. *Machine Learning* 24(1). 49–64.

Breiman, Leo. 2001. Random forests. *Machine Learning* 45(1). 5–32.

Brier, Glenn W. 1950. Verification of forecasts expressed in terms of probability. *Monthly Weather Review* 78(1). 1–3.

Bromley, Jane, Isabelle Guyon, Yann LeCun, Eduard Säckinger and Roopak Shah. 1994. Signature verification using a "Siamese" time delay neural network. In *Advances in neural information processing systems 6*.

Chen, Tianqi and Carlos Guestrin. 2016. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, 785–794. ACM.

Cho, Kyunghyun, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk and Yoshua Bengio. 2014. Learning phrase representations using RNN encoder-decoder for statistical machine translation. *arXiv preprint arXiv:1406.1078* .

Dempster, A. P., N. M. Laird and D. B. Rubin. 1977. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)* 39(1). 1–38.

Efron, Bradley. 1979. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics* 7(1). 1–26.

Elo, Arpad E. 1978. *The rating of chess players, past and present*. New York: Arco Publishing.

Friedman, Jerome H. 2001. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics* 29(5). 1189–1232.

Glickman, Mark E. 1999. Parameter estimation in large dynamic paired comparison experiments. *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 48(3). 377–394.

Glickman, Mark E. 2012. Example of the Glicko-2 system. Tech. rep., Boston University.

Guo, Chuan, Geoff Pleiss, Yu Sun and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *International conference on machine learning*, 1321–1330.

Hastie, Trevor, Robert Tibshirani and Jerome Friedman. 2009. *The elements of statistical learning: Data mining, inference, and prediction*. New York: Springer, 2nd edn.

Hinton, Geoffrey E. and Ruslan R. Salakhutdinov. 2006. Reducing the dimensionality of data with neural networks. *Science* 313(5786). 504–507.

Hitkul, Karan Aggarwal, Nishtha Yadav and Maheshkumar Dwivedy. 2019. A comparative study of machine learning algorithms for prior prediction of UFC fights. In *Harmony search and nature inspired optimization algorithms*, vol. 741, Advances in Intelligent Systems and Computing, 67–76. Singapore: Springer. doi:10.1007/978-981-13-0761-4_7.

Kaggle contributors. 2024. UFC betting odds (multi-book historical). Kaggle. Community-maintained historical UFC betting odds dump containing both moneyline rows and six-way method-of-victory props (KO/SUB/DEC for each corner); accessed in January 2025. Filtered in notebook 14 to US books and one row per `Fight_Id` (latest `adding_date`). Both moneyline and method markets used in this thesis come from this same file.

Ke, Guolin, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye and Tie-Yan Liu. 2017. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in neural information processing systems 30*.

Kingma, Diederik P. and Jimmy Ba. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* .

Niculescu-Mizil, Alexandru and Rich Caruana. 2005. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd international conference on machine learning*.

Paszke, Adam, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga et al. 2019. PyTorch: An imperative style, high-performance deep learning library. In *Advances in neural information processing systems 32*.

Pedregosa, Fabian, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot and Édouard Duchesnay. 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research* 12. 2825–2830.

Rousseeuw, Peter J. 1987. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics* 20. 53–65.

Schwarz, Gideon. 1978. Estimating the dimension of a model. *The Annals of Statistics* 6(2). 461–464.

Wolpert, David H. 1992. Stacked generalization. *Neural Networks* 5(2). 241–259.

Yan, Sheng, Linjun Liu and Comite Ubaldo. 2024. Artificial intelligence in UFC out-

come prediction and fighter strategies optimization. In *Proceedings of the 9th international conference on intelligent information processing (iciip)*, 96–100. ACM. doi:10.1145/3696952.3696966.

Yin, Jiajie. 2024. Data-driven MMA outcome prediction enhanced by fighter styles: A machine learning approach. In *Proceedings of the 4th international conference on machine learning and intelligent systems engineering (mlise)*.

Zadrozny, Bianca and Charles Elkan. 2002. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the 8th acm sigkdd international conference on knowledge discovery and data mining*, 694–699.